

# MAEG4060 Virtual Reality Systems and Applications

## Course Project — User Guide

Interactive Physical Simulation of Off-Road Vehicle Dynamics

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**About this guide.** This PDF explains how to **launch** the DirectX 9 sandbox, what **hardware** is expected, how **keyboard and gamepad** inputs map to the vehicle, and how to fix common **runtime issues**. Build-from-source and CI notes are grouped at the end for developers.

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## 1 Launch the application

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### 1.1 Game build (markers and end users)

- Open the `Game_Build` folder from your submission package (or use a folder produced by `cmake -install / scripts/package_game_build.bat`).
- **Double-click** `maeg4060_stage2.exe`. Keep the executable in the same directory as the bundled `Assets` tree.

**Tip.** If meshes or textures fail to load, you are almost certainly running the `.exe` with the wrong working directory. Launch from the folder that contains both `maeg4060_stage2.exe` and `Assets/`, or confirm that relative paths such as `Assets/models/...` resolve. Optional replacement OBJ/MTL props live under `Assets/models/replacements/` (see `README.md` there).

## 2 Hardware and software

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- **Platform:** 64-bit **Windows** (recommended).
- **Graphics:** **DirectX 9.0c** rendering path; GPU and drivers should support D3D9. Install the **DirectX 9 End-User Runtime** (or ensure equivalent system components) if you see missing-D3D errors.
- **Display:** At least **1280×720**; the window can be resized.
- **Input (optional):** **XInput-compatible** gamepad (e.g. Xbox controller) for analog throttle and steering. Keyboard control works without a pad.



**Figure 1:** *Example window: terrain splats, vehicle, props, windmill, and HUD.*

## 3 Controls

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### 3.1 Keyboard

| Key   | Action                                    |
|-------|-------------------------------------------|
| W / S | Forward / reverse (throttle)              |
| A / D | Steer left / right                        |
| Space | Brake (or hand brake, depending on build) |
| Esc   | Exit the application                      |

Bindings follow the project implementation (`GetAsyncKeyState`). If your package includes on-screen hints, treat those as authoritative for that build.

### 3.2 Gamepad (XInput)

| Control       | Action                  |
|---------------|-------------------------|
| Left stick    | Steering                |
| Right trigger | Throttle (accelerate)   |
| Left trigger  | Brake / partial reverse |

Triggers approximate analog throttle; keyboard W/S remain digital.

### 3.3 Windmill behaviour (default)

The windmill **plays automatically by default**. If the shipped windmill asset contains keyframed animation data, the hierarchy animation is advanced each frame; otherwise the blades use a simple code-driven spin for the static mesh path.

## 4 Troubleshooting

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- **Missing DLLs / DirectX errors:** Repair or install the DirectX 9 end-user runtime; update the GPU driver.
- **Black screen or device creation failure:** Prefer windowed operation first, close other exclusive full-screen apps, then retry after a driver update.
- **Assets not found:** Run from the folder layout in Section 1; verify `Assets/` sits beside the executable.
- **Gamepad ignored:** Prefer a wired connection or official wireless adapter; confirm Windows lists the device as XInput.

## 5 Build from source (developers)

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The CMake target `maeg4060_stage2` expects legacy **D3DX** headers and import libraries (`d3dx9.h`, `d3dx9math.h`, ...), which the Windows 10/11 SDK alone does not provide. Configuration uses either:

- **June 2010 DirectX SDK:** set `DXSDK_DIR` to an installation whose `Include/` and `Lib/` trees CMake can consume; *or*
- **NuGet D3DX:** leave `DXSDK_DIR` unset and allow CMake to fetch `Microsoft.DXSDK.D3DX` on first configure (requires network; first configure may take several minutes).

**Suggested command line** (matches `scripts/configure_build_release.bat`):

Run `vcvars64.bat` from your Visual Studio install, then from the repository root:

```
cmake -B build -G Ninja -DCMAKE_BUILD_TYPE=Release ^
      -DCMAKE_CXX_COMPILER=cl -DCMAKE_C_COMPILER=cl
cmake --build build
```

Ninja with an explicit `-DCMAKE_CXX_COMPILER=cl` avoids some generator detection issues seen with the Visual Studio CMake workflow. If you switch generators, delete the `build` directory (or wipe `CMakeCache.txt` and `CMakeFiles/`) before reconfiguring.

If the compiler reports **C1060** (compiler out of heap space), rebuild with less parallelism, e.g. `cmake --build build -j 1`.

**Alternative (Visual Studio generator):**

```
cmake -B build -G "Visual Studio 17 2022" -A x64
cmake --build build --config Release
```

Use this when the toolset is detected correctly.

**Package output:** run `scripts/package_game_build.bat`, or `cmake -install build -prefix Game_Build`, to populate `Game_Build/`.

## 6 GitHub Actions (optional)

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If enabled on your fork, `.github/workflows/windows-build.yml` builds on `windows-latest` and publishes a zip containing `maeg4060_stage2.exe`, `D3DX9_43.dll` when the NuGet D3DX path is used, and a copy of `Assets/`. Unzip on Windows and run the executable from that folder with `Assets` alongside it.